Jungang Zou

Department of Biostatistics, Columbia University
722 West 168th Street, New York, NY 10032
jz3183@cumc.columbia.edu Website Google Scholar GitHub

Research Interests: Bayesian modeling; data integration; AI for statistics; causal inference; missing data; survey analysis.

Education

09/2022–Present	**Columbia University** Ph.D. in Biostatistics, expected May 2027 Dissertation: *Data Integration Using Bayesian Machine Learning Methods* Advisor: Dr. Qixuan Chen	New York, NY
09/2019–05/2022	**Columbia University** M.S. in Biostatistics Gap year: 2020–2021 due to COVID-19	New York, NY
09/2015–06/2019	**Xiamen University** B.E. in Computer Software Engineering	Xiamen, China

Awards and Honors

2025 Travel Award, 14th International Conference on Bayesian Nonparametrics.
2026 Travel Award, IMSI Workshop: New Horizons on Model Transportability and Data Integration.
2026 Outstanding Doctoral Teaching Assistant Award, Columbia University's Department of Biostatistics.
2026 Student Paper Competition Award, ASA GSS/SRMS/SSS Sections, JSM 2026.
2026 Paper Award, STAI-X 2026 (one of 9 awards selected from all 54 accepted papers).

Publications (Google Scholar)

Statistical Methodology

[1] Hu, L., **Zou, J.**, Gu, C., Ji, J., Lopez, M. & Kale, M. A Flexible Sensitivity Analysis Approach for Unmeasured Confounding with Multiple Treatments and a Binary Outcome with Application to SEER-Medicare Lung Cancer Data. *The Annals of Applied Statistics* **16,** 1014–1037 (2022).

[2] Williams, S. Z., **Zou, J.**, Liu, Y., Si, Y., Galea, S. & Chen, Q. Improving Survey Inference Using Administrative Records without Releasing Individual-Level Continuous Data. *Statistics in Medicine* **43,** 5803–5813 (2024).

[3] Zhang, K., Li, Y., **Zou, J.**, Wang, S., Fausto, B. A. & Hu, L. Multi-Level Variable Selection Using a BART-Enhanced Mixed-Effects Framework. *Statistics in Medicine* **45,** e70593 (2026).

[4] **Zou, J.**, Wang, S. & Chen, Q. Bayesian MI-LASSO for Variable Selection on Multiply-Imputed Data. arXiv: 2211.00114 [stat.ME] (2026+). Minor revision, *Statistics in Medicine*.

[5] **Zou, J.** & Hu, L. Nonparametric Bayesian Additive Regression Trees for Prediction and Missing Data Imputation in Longitudinal Studies. (2026+). Major revision, *Computational Statistics and Data Analysis*.

[6] **Zou, J.**, Jiang, A. Z. & Chen, Q. AI4BayesCode: From Natural Language Descriptions to Validated Modular Stateful Bayesian Samplers. arXiv: 2605.18476 [stat.CO] (2026+). Submitted to *Journal of the American Statistical Association* via the STAI-X 2026 fast-track review invitation.

[7] **Zou, J.**, Schwartz, J. E., Little, R., Moise, N. & Chen, Q. Bayesian Propensity Score Predictive Inference for Extending Trial Findings to Target Populations. (2026+). Submitted to *The Annals of Applied Statistics*.

[8] **Zou, J.**, Lin, F., Gao, S., Deng, G., Zeng, W. & Alterovitz, G. Transfer Learning Based Multi-Objective Genetic Algorithm for Dynamic Community Detection. arXiv: 2109.15136 [cs.SI]. 20+ citations on Google Scholar.

[9] **Zou, J.**, Jiang, A. Z. & Chen, Q. Bayesian Federated Nonlinear Variable Selection under Site Heterogeneity with Unknown Sharing Structure. (2026+). In preparation.

[10] Wei, Y., **Zou, J.** & Chen, Q. MetaBART: Bayesian Additive Regression Trees for Meta-Regression and Generalization from Aggregate Data to Target Populations. (2026+). In preparation.

[11] **Zou, J.**, Zou, Y. & Hu, L. A Bayesian Iterative Conditional Expectation G-Formula for Causal Survival Analysis under Dynamic Treatment Strategies. (2027+). In preparation.

Collaborative Research

[1] Jadow, B. M., Hu, L., **Zou, J.**, Labovitz, D., Ibeh, C., Ovbiagele, B. & Esenwa, C. Historical Redlining, Social Determinants of Health, and Stroke Prevalence in Communities in New York City. *JAMA Network Open* **6,** e235875 (2023).

[2] Flores, N. M., Lovinsky-Desir, S., Divjan, A., Hoepner, L. A., **Zou, J.**, Miller, R. L., Herbstman, J. B., Perera, F. P., Perzanowski, M. S. & Chen, Q. Trajectory Analysis of Rhinitis in a Birth Cohort from Lower-Income New York City Neighborhoods. *Journal of Allergy and Clinical Immunology* **154,** 111–119 (2024).

[3] Greenleaf, A. R., Francis, S., **Zou, J.**, Farley, S. M., Lekhela, T., Asiimwe, F. & Chen, Q. Influenza-Like Illness in Lesotho from July 2020 to July 2021: Population-Based Participatory Surveillance Results. *JMIR Public Health and Surveillance* **10,** e55208 (2024).

[4] Onyilofor, C., Feng, B., **Zou, J.**, Mizhquiri Barbecho, J., Duran, A. T., Ye, S., Garcia, O., Singer, J., Chen, Q. & Moise, N. Assessing the Fidelity of Depression Screening Implementation in Coronary Heart Disease Patients: A Cross-Sectional Study. *Journal of the American Heart Association* **13,** e035550 (2024).

[5] Xu, Z., Scharp, D., Hobensack, M., Ye, J., **Zou, J.**, Ding, S., Shang, J. & Topaz, M. Machine Learning-Based Infection Diagnostic and Prognostic Models in Post-Acute Care Settings: A Systematic Review. *Journal of the American Medical Informatics Association* **32,** 241–252 (2025).

[6] Flores, N. M., Casey, J. A., Lovinsky-Desir, S., Spira-Cohen, A., Louis, S., Walters, S., Little, M., **Zou, J.**, Rudolph, K. E., Hernández, D., Chen, Q. & Perzanowski, M. S. Impact of a Large-Scale Mold Intervention in Public Housing on Asthma Emergency Department Visits. *American Journal of Respiratory and Critical Care Medicine*, aamag441 (2026).

[7] Kim, K., **Zou, J.**, Feng, B., Luan, M., Jones, A., Gadbois, D., Lim, J., Schwartz, J. E., Chen, Q. & Moise, N. Predictors of Sustainability in the Collaborative Care Medicaid Program for Depression: A Cross-Sectional Study. *Psychiatric Services* **77,** 718–724 (2026).

Tools and Software

[1] AI4BayesCode: From Natural Language Descriptions to Validated Modular Stateful Bayesian Samplers. (Website | GitHub).

[2] PSPI: Propensity Score Predictive Inference for Generalizability and Transportability (R | GitHub).

[3] SBMTrees: Sequential Imputation with Bayesian Trees Mixed-Effects Models for Longitudinal Data (R | GitHub).

[4] AuxSurvey: Survey Analysis with Auxiliary Discretized Variables (R | GitHub).

[5] SAMTx: Sensitivity Assessment to Unmeasured Confounding with Multiple Treatments (R).

[6] BMIselect: Bayesian MI-LASSO Models for Variable Selection on Multiply-Imputed Data (R | GitHub).

[7] TMOGA: A Multi-Objective Genetic Algorithm for Dynamic Community Detection (PyPI | GitHub).

Presentations

Invited Presentations

[1] **Zou, J.** *Nonparametric Bayesian Additive Regression Trees for Prediction and Missing Data Imputation in Longitudinal Studies.* ENAR 2023 Spring Meeting (Nashville, TN, USA). Invited session: *Bayesian Machine Learning for Decision-Making with Incomplete Information*. Mar. 2023.

[2] **Zou, J.** *Bayesian Propensity Score Predictive Inference for Extending Trial Findings to Target Populations.* SILVER Lab Meeting, Department of Epidemiology and Public Health, University of Maryland School of Medicine. Aug. 2026.

[3] **Zou, J.** *AI4BayesCode: From Natural Language Descriptions to Validated Modular Stateful Bayesian Samplers.* Technology and Artificial Intelligence Learning Seminar (TAILS), Montana State University. Forthcoming. Dec. 2026.

Contributed Talks

[1] **Zou, J.** *Generalizing Treatment Effects from Trials to EHR Populations Using Propensity Score Predictive Inference.* ENAR 2026 Spring Meeting (Indianapolis, IN, USA). Mar. 2026.

[2] **Zou, J.** *AI4BayesCode: From Natural Language Descriptions to Validated Modular Stateful Bayesian Samplers.* STAI-X 2026 (Boston, MA, USA). Aug. 2026.

[3] **Zou, J.** *Bayesian MI-LASSO for Variable Selection on Multiply-Imputed Data: Application to an Environmental Health Survey.* JSM 2026 (Boston, MA, USA). Aug. 2026.

Posters

[1] **Zou, J.**, Wang, S. & Chen, Q. *Variable Selection for Multiply-Imputed Data: A Bayesian Framework.* 2023 Columbia Biostatistics Annual Research Symposium (New York, NY, USA). Sept. 2023.

[2] **Zou, J.**, Schwartz, J. E., Little, R., Moise, N. & Chen, Q. *Generalizing Treatment Effects from EHR-Recruited Trials Using Bayesian Propensity Prediction.* 2025 Columbia Biostatistics Annual Research Symposium (New York, NY, USA). Apr. 2025.

[3] **Zou, J.** & Hu, L. *Nonparametric Bayesian Additive Regression Trees for Prediction and Missing Data Imputation in Longitudinal Studies.* 14th International Conference on Bayesian Nonparametrics (Los Angeles, CA, USA). June 2025.

[4] **Zou, J.**, Schwartz, J. E., Little, R., Moise, N. & Chen, Q. *Transporting Treatment Effects from Randomized Trials to EHR Populations Using Bayesian Propensity Score Predictive Inference.* IMSI Workshop: New Horizons on Model Transportability and Data Integration (Chicago, IL, USA). June 2026.

[5] **Zou, J.**, Jiang, A. Z. & Chen, Q. *AI4BayesCode: From Natural Language Descriptions to Validated Modular Stateful Bayesian Samplers.* STAI-X 2026 (Boston, MA, USA). Aug. 2026.

Teaching

Teaching Assistant, Columbia University

Fall 2021, 2023, 2025	Analysis of Health Surveys (P8123)
Fall 2022	Applied Regression II (P8110)
Spring 2023	Statistical Inference (P8109)
Summer 2023, 2024	Statistical Analysis with Missing Data Workshop
Spring 2024	Topics in Advanced Statistical Computing (P8160)
Fall 2024, 2026	Clinical Trial Methodology (P8142)
Spring 2025, 2026	Data Science II (P8106)

Academic Service

2026 Session Chair, *Sampling, Weighting, Missing Data and Variance Estimation*, JSM 2026.
2026 Reviewer, *Computational Statistics*.

Professional Memberships

American Statistical Association (ASA)
Eastern North American Region of the International Biometric Society (ENAR)
International Chinese Statistical Association (ICSA)
Institute of Mathematical Statistics (IMS)
International Society for Bayesian Analysis (ISBA)

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